Systematic Trading Strategies with Machine Learning Algorithms

Introduction to Systematic Strategies with ML

April 21, 2026

Outline

Course Syllabus Overview

Total: 9 Lectures

- Every lecture includes a programming session.
- 2 project sessions (no theory): Lectures 5 and 8.
- Weekly office hours: **Wednesdays, 1pm–2pm and 5pm–6pm**.
- Fridays 4pm–5pm: optional session (Weeks 2, 5, 8) or additional office hours.
- Fridays 5pm–6pm: additional office hours on weeks with an optional session.

Syllabus Breakdown

Lecture	Theory	Programming Session
Lecture 1: *Introduction to Systematic Strategies with Machine Learning*	Introducing systematic strategies using machine learning, covering labeling methods, evaluation metrics, feature importance analysis, and hyperparameter optimization for sequential data. Introducing the Meta-labeling approach, which balances recall and precision by identifying conditions under which a primary model yields optimal trading performance.	Practical session on financial data labeling for trend detection.
Lecture 2: *Unsupervised Learning and Clustering Techniques*	Introducing clustering algorithms and dimensionality reduction techniques, highlighting their role in addressing substitution effects in feature importance analysis.	Improving model performance by reducing substitution effects among correlated features using cluster-level feature importance analysis.
Lecture 3: *Latent Variable Models in Financial Asset Regime Detection*	Introducing Hidden Markov Models (HMMs) for predicting financial market regimes.	Implementation of the HMM in a simple case of discrete observations.
Lecture 4: *Supervised Learning Algorithms*	Overview of key supervised learning algorithms with emphasis on ensemble models (Bagging and Boosting) and feature importance analysis. Introducing Gated Residual Networks (GRN) and Variable Selection Networks (VSN).	Forecasting time series with gradient boosting.
Lecture 5: *Project: Enhancing Strategy Performance in Crypto Markets*	No theory.	Applying concepts from previous sessions to filter out non-profitable trades from a primary model on crypto data, using a tree-based secondary model with an emphasis on cluster-level feature importance analysis.

Syllabus Breakdown (cont.)

Lecture	Theory	Programming Session
Lecture 6: *Introducing Variable Selection Networks*	Introducing the Variable Selection Network (VSN) from the Temporal Fusion Transformer architecture, covering its role in handling mixed-type inputs and selecting relevant features at each time step.	Implementing a specialised neural network architecture handling numerical and categorical data through embedding techniques and Gated Residual Networks, with feature-level interpretability.
Lecture 7: *Neural Networks for Interpretable Time Series Forecasting*	Exploration of neural network models for interpretable time series forecasting, including GRUs, LSTMs, N-Beats and Transformers.	Coding a Transformer from scratch.
Lecture 8: *Project: Volatility Forecasting with Temporal Fusion Transformers*	No theory.	In-depth exploration of sequential neural network architectures for predicting volatility, replicating the Temporal Fusion Transformer study and evaluating it against earlier models.
Lecture 9: *Review*	Mock exam and revision elements to prepare for the final exam.	No programming session.

Optional Sessions

Scheduled on Fridays 4pm–5pm via Zoom. Not required for evaluation, but highly relevant for preparation and industry applications.

- **Optional Session 1 (April 24):** Probability and Calculus Refresher
  Preparation for Session 3: Latent Variable Models and HMMs
- **Optional Session 2 (May 15):** Introduction to Large Language Models
  From next-word prediction to fine-tuning
- **Optional Session 3 (June 5):** Building Reasoning Models
  Chain-of-Thought, GRPO, DeepSeek-R1, and distillation

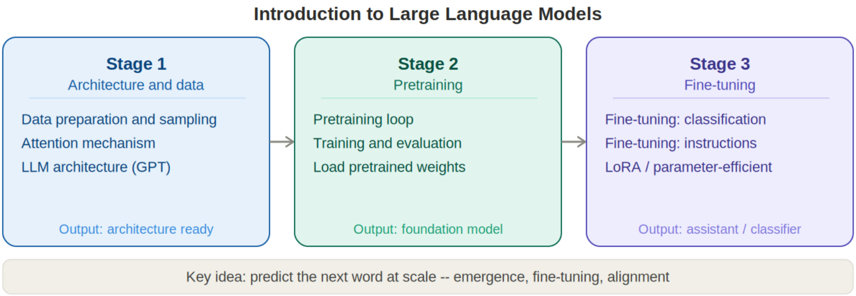
Optional Session 1

Probability and Calculus Refresher

Probability	Matrix calculus	Stat. models and MLE
Sum and product rules	Gradient and Hessian	Bernoulli, Binomial
Cond. independence	Linear/quadratic forms	Multinomial, Gaussian
Bayes formula	Differentials, chain rule	Maximum likelihood (MLE)

This session covers the mathematical foundations needed for Session 3. We review the key rules of probability, introduce matrix calculus tools used in optimisation, and cover the main statistical models alongside maximum likelihood estimation.

Optional Session 2



We walk through the three stages of building an LLM from scratch: implementing the architecture and data pipeline, pretraining on unlabeled text to obtain a foundation model, and fine-tuning it for classification or instruction following.

We cover the three complementary approaches to making LLMs reason: spending more compute at inference time, training with reinforcement learning and verifiable rewards, and distilling reasoning capabilities from a large teacher into a smaller model.

Evaluation and Final Project

Your final grade will be based on:

- **50% Final Exam**
  Standard written evaluation based on the course content.
- **50% Challenge with Alken Asset Management**
  You will work in teams on a machine learning challenge applied to real-world financial data, in collaboration with Alken's portfolio management team.

ALKEN

MCQs and Office Hours

- After each lecture, you will find a short set of MCQs on the course platform.
- These are **optional**, not graded, and meant to test your understanding of key concepts.
- They can also serve as a self-check before tackling the project or exam.

Need help?
Feel free to send me an email anytime. I will try to respond as quickly as possible.

I highly encourage you to attend office hours every Wednesday from 1pm–2pm and 5pm–6pm.

Outline

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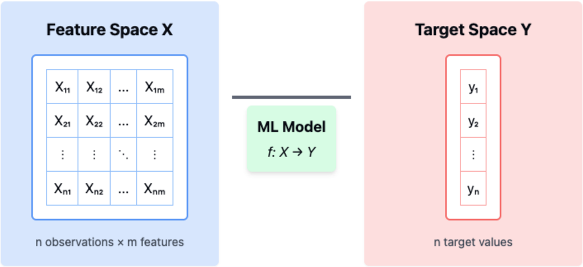
Supervised Learning:

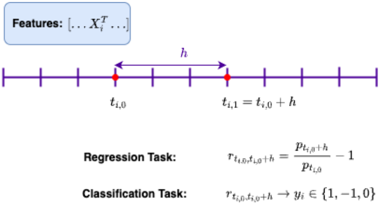
- Learn a function $f : X \to Y$ from labeled data.
- **Feature space** X: matrix of features, n observations $\times$ m features.
- **Target space** Y: vector of n labels (or target values).



Regression vs Classification

Supervised Learning Tasks:

- **Regression:**
  - Predict a continuous outcome.
  - $y_i \in \mathbb{R}$, e.g., temperature, price, return value.
- **Classification:**
  - Predict a discrete label from a finite set.
  - $y_i \in \{c_1, c_2, \ldots, c_K\}$, e.g., up/down, buy/sell/hold.



Labeling: Why It Matters

Labeling defines the task that the ML algorithm learns — it encodes the investment hypothesis.

- The choice of label determines what features are informative.
- Features that are irrelevant for one labeling scheme may be predictive under another.
- Understanding feature relevance requires proper **Feature Importance Analysis** (see Lecture 2).

Labeling Methods covered in the course:

- **Fixed-Time Horizon Labeling** return over a fixed *h* period.
- **Triple Barrier Labeling** incorporates profit-taking, stop-loss, and timeouts.
- **Trend Scanning** detects regime changes (Programming Session 1).

Fixed-Time Horizon Labeling

Most common method in financial ML papers.

$$y_i = \begin{cases} -1 & \text{if } r_{t_{i,0},t_{i,0}+h} < -\tau \\ 0 & \text{if } |r_{t_{i,0},t_{i,0}+h}| \leq \tau \\ 1 & \text{if } r_{t_{i,0},t_{i,0}+h} > \tau \end{cases} \quad \text{with} \quad r_{t_{i,0},t_{i,0}+h} = \frac{p_{t_{i,0}+h}}{p_{t_{i,0}}} - 1$$

- h: number time bars.
- τ: return threshold.

Problems:

- Same threshold τ used regardless of volatility.
- Ignores price path, only cares about return at $t + h$.

Fixed-Time Horizon: Pitfalls

Why this method is often flawed:

- Labeling does not account for volatility: a 1% move in low-volatility vs. high-volatility regimes is treated the same.
- It ignores stop-losses and the path of returns, key to real trading decisions.

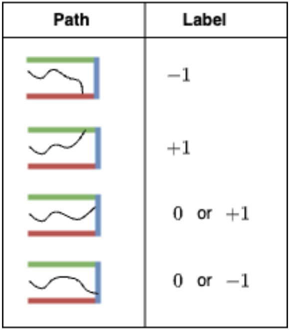
Alternatives:

- Use volatility-adjusted thresholds.
- Adopt path-aware methods (next slide).

The Triple Barrier Method

A more realistic approach based on trading logic.

- Label based on which of three events occurs first:
  1. Hits **stop-loss** barrier $\Rightarrow y_i = -1$
  2. Hits **profit-taking** barrier $\Rightarrow y_i = +1$
  3. **Time-out** at horizon h
     - $\Rightarrow y_i = 0$, or sign of return
- *Encodes risk management logic directly into the labeling process.*



Triple Barrier: Implementation Details

- Width of barriers can be based on predicted volatility over the interval.
- Records the time $t_{i,1}$ of the first barrier hit.

Advantage: Path-aware labels depend on what happens during the position.

Caveat: Barrier crossing is a discrete event and may miss label transitions by a small margin.

Alternative: The caveat is addressed by the following method.

Labeling by Detecting Trends (Trend Scanning)

Trend Scanning: label based on observed trends without explicit barriers or fixed horizon h.

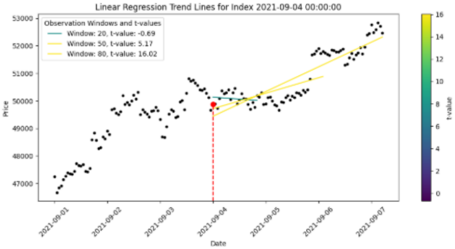
- Fit a linear model to a forward-looking window of length H:

$$x_{t+h} = \beta_0 + \beta_1 h + \varepsilon_{t+h}, \quad h = 0, \ldots, H-1 \quad \Rightarrow \quad \hat{t}_{\widehat{\beta}_1} = \frac{\widehat{\beta}_1}{\widehat{\sigma}_{\widehat{\beta}_1}}$$

- H is the **look-forward period**.
- Different values of H produce different t-statistics.
- We select the H that maximizes $|\hat{t}_{\widehat{\beta}_1}|$.
- Label is assigned based on the sign and magnitude of the most significant slope.

Trend Scanning in Practice

- Trend Scanning can also be used backward in time to generate **features** indicating historical trend strength.
- This method is data-driven and adaptive, it avoids arbitrary thresholds.
- **Programming Session 1:** Trend Scanning Implementation.



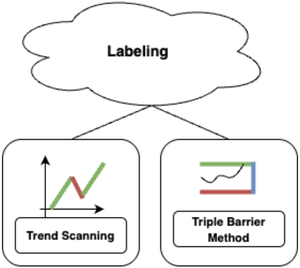
Labeling: Summary of Methods

Fixed-Time Horizon:

- Simplest method: return over a fixed period h.
- Ignores path-dependence, volatility, and trading constraints.

Path-Aware Methods:

- **Triple Barrier:** encodes risk management — profit-taking, stop-loss, and max-hold time.
- **Trend Scanning:** selects statistically significant trend based on forward regression.



Click here to take the quiz

Outline

Evaluation Metrics – Notation

General Setup:

- N: number of samples, L: number of classes
- $y_i \in \{1, \ldots, L\}$: true label for sample i
- $\hat{y}_i \in \{1, \ldots, L\}$: predicted label for sample i
- $\hat{\mathbf{s}}_i = f(\mathbf{x_i}) \in \mathbb{R}^L$: predicted score vector.
- $\hat{s}_{i,\ell}$: predicted score for class ℓ on sample i
- **Predicted Scores** ($\hat{s}_{i,\ell}$) are used for probabilistic evaluation (e.g., log loss, AUC).
- **Predicted Labels** are computed via:
  - **Multiclass:** $\hat{y}_i = \arg\max_\ell \hat{s}_{i,\ell}$
  - **Binary:** $\hat{y}_i = \mathbb{1}[\hat{s}_{i,1} > \tau]$, with threshold $\tau \in [0, 1]$

Evaluation Metrics – Accuracy

Accuracy Score measures the proportion of correct predictions:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{y}_i = y_i]$$

- Simple and intuitive.
- Works best for balanced classification problems.
- Can be misleading for imbalanced classes.

Evaluation Metrics – Confusion Matrix

Confusion Matrix (Binary Classification):

	Predicted Positive	**Predicted Negative**
Actual Positive	TP	FN
Actual Negative	FP	TN

- TP: True Positives — correctly predicted positives
- FP: False Positives — incorrectly predicted positives
- FN: False Negatives — missed actual positives
- TN: True Negatives — correctly predicted negatives

Evaluation Metrics – Precision and Recall

Precision: How many of the predicted positives are actually positive?

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\text{TP}}{\text{Predicted Positive}}$$

Recall: How many of the actual positives did we correctly identify?

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{\text{TP}}{\text{Actual Positive}}$$

F1 Score: Harmonic mean of precision and recall

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when dealing with class imbalance or cost-sensitive errors.

Evaluation Metrics – Logarithmic Loss (LogLoss)

Binary Classification:

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log(\hat{s}_i) + (1 - y_i)\log(1 - \hat{s}_i)$$

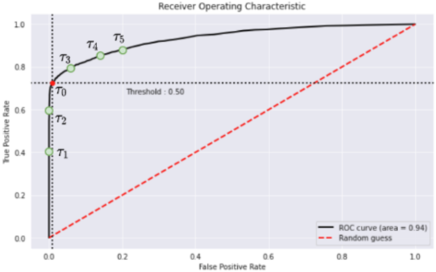
Multiclass:

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{l=1}^{L} y_{il}\log(\hat{s}_{il})$$

Uses predicted scores, not just final decision. Strongly penalizes high confidence on wrong class.

ROC (Receiver Operating Characteristic) Curve:

- Plots **True Positive Rate (TPR)** vs. **False Positive Rate (FPR)**.
- TPR (Sensitivity): $\frac{TP}{TP+FN}$
- FPR: $\frac{FP}{FP+TN}$
- Each point corresponds to a different threshold on predicted scores.

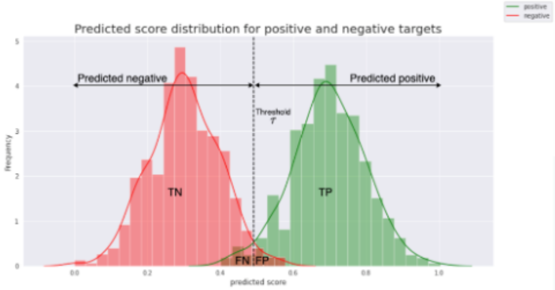


Threshold dependence: Changing the threshold traces the full ROC curve.

Evaluation Metrics – AUC (Area Under Curve)

AUC (Area Under ROC Curve):

- **Interpretation:** The probability that a randomly selected positive example has a higher score than a randomly selected negative one. (Optional: Click *here* to understand why both definitions are equivalent.)
- Threshold-independent: considers all possible classification thresholds.



Evaluation Metrics – Summary

Key Metrics to Evaluate Classification Models:

- **Accuracy** – Proportion of correct predictions. Works best with balanced classes.
- **Precision & Recall** – Precision: TP / Predicted Positive. Recall: TP / Actual Positive.
- **F1 Score** – Harmonic mean of precision and recall. Useful in imbalanced settings.
- **Log Loss** – Uses predicted scores. Penalizes overconfidence in wrong predictions.
- **ROC Curve** – Plots TPR vs. FPR for all thresholds.
- **AUC** – Probability that a randomly chosen positive is ranked above a negative.

Outline

Hyperparameter Tuning with Cross-Validation

Goal: Find the best set of hyperparameters $h \in \mathcal{H}$ that generalizes well.

- For each hyperparameter configuration $h \in \mathcal{H}$, we evaluate the model using K-fold cross-validation.
- Performance is measured using an evaluation metric $\mathcal{E}$, e.g., accuracy, log-loss, AUC.
- Select $h^* = \arg\max_{h \in \mathcal{H}} \mathcal{E}^h$

Example (Tree-Based Models):

- $\mathcal{H} = \{\text{max depth}, \text{min samples split}, \text{criterion}, \dots\}$
- Use CV to compare scores across $h \in \mathcal{H}$

Cross-Validation for Hyperparameter Selection

Algorithm Cross-Validation for Hyperparameter Tuning

Require: Dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, hyperparameter grid $\mathcal{H}$, evaluation metric $\mathcal{E}$, number of folds K

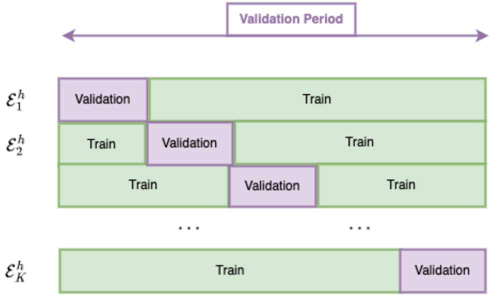
Ensure: Best hyperparameter h^*

```
for each h ∈ H do
    Initialize E^h ← 0
    for each fold k = 1 to K do
        Split data into training set D_train^(k), test set D_test^(k)
        Train model with h on D_train^(k)
        Compute score E_k^h ← E(D_test^(k))
    end for
    Average score: E^h ← (1/K) Σ_{k=1}^{K} E_k^h
end for
return h* = arg max_{h∈H} E^h
```

- **for** each $h \in \mathcal{H}$ **do**
  - Initialize $\mathcal{E}^h \leftarrow 0$
  - **for** each fold $k = 1$ to K **do**
    - Split data into training set $D_{\text{train}}^{(k)}$, test set $D_{\text{test}}^{(k)}$
    - Train model with h on $D_{\text{train}}^{(k)}$
    - Compute score $\mathcal{E}_k^h \leftarrow \mathcal{E}(D_{\text{test}}^{(k)})$
  - **end for**
  - Average score: $\mathcal{E}^h \leftarrow \frac{1}{K} \sum_{k=1}^{K} \mathcal{E}_k^h$
- **end for**
- **return** $h^* = \arg\max_{h \in \mathcal{H}} \mathcal{E}^h$

K-Fold Cross-Validation:

- Data is split into K folds.
- Each fold is used once as a test set, remaining as training.
- Final score is the average across all folds.



$$h \in \mathcal{H} \quad \to \quad \mathcal{E}^h = \frac{1}{K}\sum^{K} \mathcal{E}_k^h$$

The Problem of Information Leakage in Finance

Standard CV fails in finance due to time-series structure:

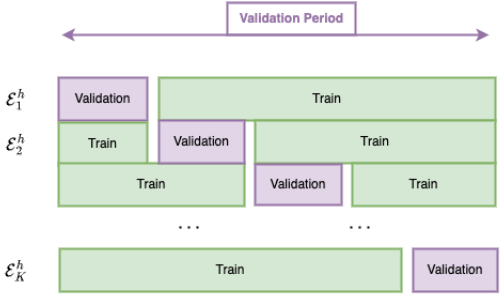
- Data is not IID, financial time series are autocorrelated.
- Overlapping labels cause **label leakage** between training and test sets.

Two key techniques from to prevent leakage:

- **Purging:** remove training observations whose label periods overlap with test labels.
- **Embargo:** exclude training data immediately following the test window to avoid serial correlation.

Purged K-Fold Cross-Validation:

- Removes overlapping label periods (purging).
- Applies a time-based buffer around test windows (embargo).
- Ensures the training set is informationally clean.



$$h \in \mathcal{H} \quad \rightarrow \quad \mathcal{E}^h = \frac{1}{K}\sum_{k=1}^{K} \mathcal{E}_k^h$$

Click here to participate in the poll

Outline

Why Do We Need Meta-Models?

A primary model — whether targeting trend detection or mean reversion — will perform well only in specific market contexts.

- Market regimes evolve: what works in one regime may fail in another.
- We want to either:
  - Adjust position sizing based on some level of confidence.
  - Filter out trades in regimes where the primary model underperforms.

Meta-labeling trains a secondary model to detect the contexts where the primary model has edge and when to trust it.

What is a Meta-Model?

Meta-Model: A secondary ML model learns when to trust the primary model.

- Introduced by Marcos López de Prado.
- Primary model predicts the trade direction.
- Metamodel decides whether to act (Trade T) or abstain ($\overline{T}$).
- Helps convert a weak signal into a strong one.

Applicable to any directional model or expert.

Metamodel Features vs Primary Model Features

Key distinction:

- The **Primary Model** is trained to detect trends or predict return direction.
- The **Metamodel** is trained to detect the conditions under which the primary model works well.

Why separate features?

- Using the same features risks learning redundant information.
- The Metamodel is not predicting the market, it is predicting the *performance of a model*.

Think of the Metamodel as a regime detector: it learns the environments where the primary model has alpha.

Meta-Labels: The Triple Barrier Method

We define a meta-label $(T, \overline{T})$ using the outcome of each trade:

- T: If the trade would have hit profit-taking before stop-loss.
- $\overline{T}$: Otherwise, trade not worth taking.

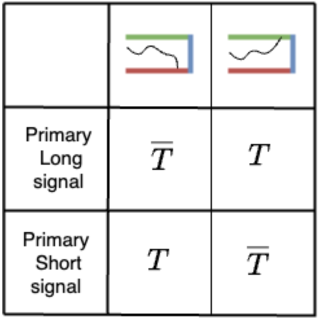


Figure: Meta-labels are derived from the result of the primary model's trade using the triple-barrier method.

Overview of the Meta-model Framework

The Metamodel takes as input the predictions of the primary model, as well as features describing market conditions, and learns to filter or size trades based on expected performance.

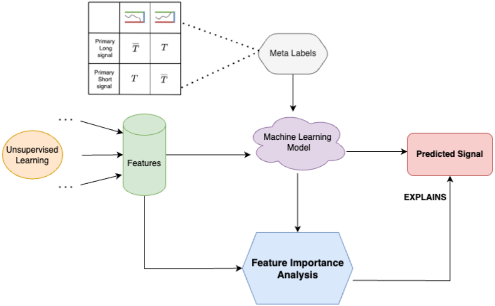


Figure: The Metamodel learns a mapping between market context and the reliability of the primary model's signal.

Click here to take the quiz

Trading Recall for Precision: Improving F1 Score

Primary model: high recall (takes every trade), low precision.

Metamodel: reduces false positives to improve precision.

- The trade-off is controlled by a decision threshold τ.
- Goal: maximize F1-score = harmonic mean of precision and recall.

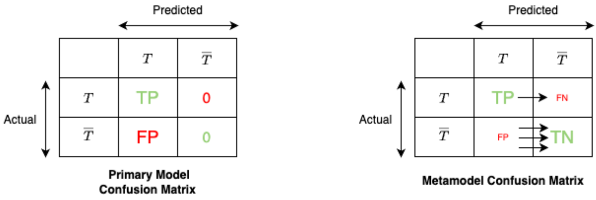


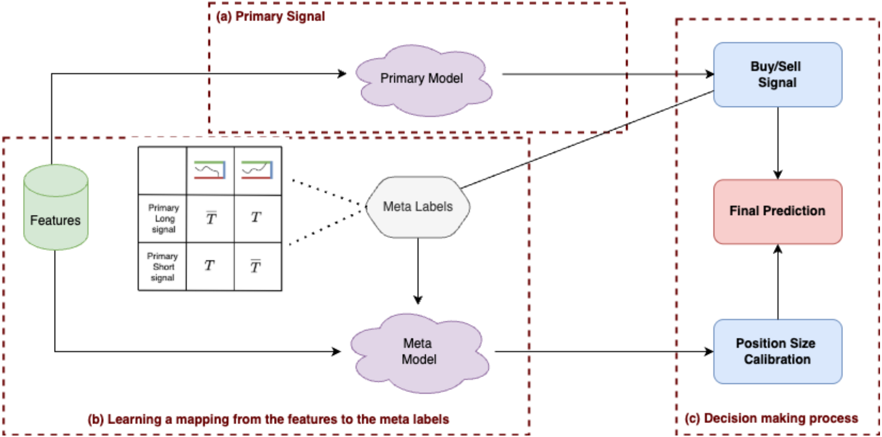
Figure: The Metamodel improves the confusion matrix by rejecting low-confidence trades.

We'll implement meta-labeling in Lecture 5 to filter out weak signals from a crypto directional model.

5 steps:

1. We have trained a primary model to predict direction (Long/Short).
2. Use the triple barrier method to label trades as T or $\overline{T}$.
3. Train a Metamodel to predict whether to take the trade.
4. Use Feature Importance Analysis to understand Market Regimes under which the primary model performs well.
5. Compare the performances of the primary model (Long/Short) with the Final Strategy (Long / Short / Neutral).

Applying Meta-Labeling to a Crypto Strategy



Feedback Poll

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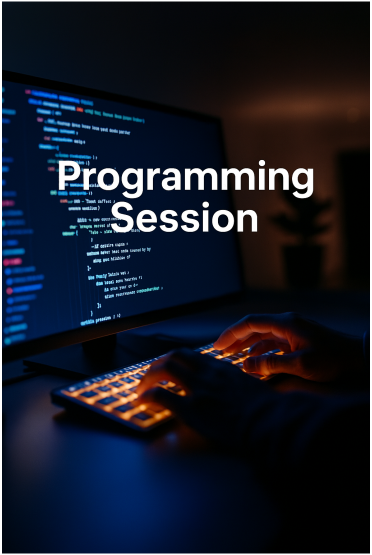

Programming Session 1: Implementation of the Trend Scanning Method.

- *Click here to access the programming session*

Solution will be posted tonight on the GitHub page.

- *Click here to access ccess the GitHub Page*

Thank you for your attention